Convergence of Trust-Adaptive Shadow Aggregation for Byzantine-Robust Federated Learning under Non-IID Data

[Author names omitted for double-blind review]
[institution]@[domain]

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Abstract

We establish convergence guarantees for Trust-Adaptive Shadow Aggregation (TASA), the aggregation mechanism at the heart of the TAVS-ESP Byzantine-robust federated learning system [**?**]. TASA introduces two features absent from all existing convergence analyses: *(i)* trust-adaptive discount weights $p_i(r) \in (0, 1)$ derived from Bayesian posterior honesty probabilities that vary over rounds according to a geometrically-grounded trust score, creating temporal correlation that invalidates standard independence assumptions; and *(ii)* a block-variance-normalized outlier detector whose false positive rate for honest non-IID clients is $O(1)$ by construction, in contrast to scalar-threshold detectors whose false positive rate grows with gradient heterogeneity ζ.

We prove convergence via a four-stage Lyapunov argument. **Stage 1** (IID, fixed weights) establishes a baseline geometric rate with an explicit shadow-aggregation bias term B_p. **Stage 2** (IID, trust-dynamic weights) introduces a composite Lyapunov potential $\Phi(r) = F(w(r)) - F^* + c\sum_i (T_i(r) - T_i^*)^2$ and proves it is a supermartingale once trust scores stabilize, yielding convergence at rate $\gamma = \min(\mu\eta,\, 1 - \alpha^2)$. **Stage 3** (non-IID, Byzantine) conditions on the Layer-2 detection event and bounds the residual Byzantine and heterogeneity bias terms separately, giving a high-probability convergence guarantee. **Stage 4** (non-convex, LLM fine-tuning) proves convergence to a stationary-point neighborhood at rate $O(1/\sqrt{T})$ plus bounded additive constants.

Our central comparison result shows that the convergence floor under block-variance-normalized detection is *strictly smaller* than under scalar-threshold detection by a factor of $\Omega(\zeta^2/\sigma_g^2)$, where ζ is gradient dissimilarity and σ_g^2 is honest gradient variance. We validate the theory on federated fine-tuning of LLaMA-3-8B under a Sleeper Agent attack, confirming that the block-variance detector maintains near-zero false positive rates at heterogeneity levels where scalar-threshold detection degrades significantly.

1 Introduction

Federated learning (FL) convergence theory has matured significantly, with tight analyses available for FedAvg [8], heterogeneous FedProx [7], and variance-reduced SCAFFOLD [5]. Byzantine-robust FL has its own convergence literature, establishing that robust aggregators converge at rates degraded by the Byzantine fraction [3, 6]. However, no existing work analyzes convergence when:

- Gradient aggregation weights are *trust-adaptive*—varying over rounds based on client history—creating temporal correlations that violate the independence assumptions in all prior proofs.
- Some clients are aggregated without verification at *discounted weights* derived from a Bayesian posterior, and the discount weights themselves evolve via an EMA of a geometric detection signal.
- The outlier detector uses *block-variance normalization* that separates structural data heterogeneity from adversarial anomalies, changing the false-positive rate as a function of heterogeneity.

This paper fills this gap. Our contributions are:

- **TC2 (four stages):** The first convergence proof for FL with trust-adaptive shadow aggregation under non-IID data and Byzantine workers. Each stage is self-contained and independently publishable.
- **Comparison theorem:** A formal result showing the convergence floor under block-variance-normalized detection is strictly tighter than under scalar-threshold detection, quantified in terms of gradient heterogeneity ζ.
- **Trust stability lemma:** A characterization of EMA trust score dynamics including the rate of convergence to steady state and its dependence on α and the variance of the behavioral score.
- **Empirical validation:** Experiment E3 on LLaMA-3-8B with LoRA under a Sleeper Agent attack, confirming the theoretical predictions about false positive rates and convergence floor at multiple heterogeneity levels.

Companion paper. TAVS-ESP system design, security theorems (TC1, TC3, TC4), Sybil resistance, and experiments E1/E2/E4/E5 are in **?**]. This paper is self-contained but builds on the system described there; we briefly recap the relevant components in Section 3.

2 Related Work

FL convergence under heterogeneous data. Li et al. [7] analyzed FedAvg with a proximal regularizer under non-IID data, establishing convergence with a gradient dissimilarity term. Karimireddy et al. [5] introduced control variates to remove client drift, achieving tighter rates. Yuan and Ma [12] analyzed partial client participation without security considerations. None of these works analyze trust-adaptive weights or Byzantine robustness.

Byzantine FL convergence. Karimireddy et al. [6] proved convergence of Byzantine SGD under the assumption of a fixed robust aggregator applied to all clients every round. Data and Diggavi [3] characterized the information-theoretic limits of Byzantine-robust gradient aggregation. Wu et al. [10] analyzed convergence with noisy Byzantine detection, but with binary include/exclude decisions rather than continuous trust-weighted aggregation. Our work differs in allowing fractional weights that vary with trust history.

Shadow aggregation and soft exclusion. Blanchard et al. [1] and subsequent robust aggregators use hard exclusion. Cao et al. [2] uses trust scores for weighted aggregation but analyzes neither convergence under trust dynamics nor the interaction with non-IID data and a varying verified set. To our knowledge, no prior work proves convergence for an aggregation rule with temporally correlated, trust-adaptive, fractional weights derived from a geometric detection signal.

Non-convex FL convergence. Ghadimi and Lan [4] established $O(1/\sqrt{T})$ convergence for non-convex SGD. Yang et al. [11] and Zhu et al. [13] extended this to FL settings. Stage 4 of our proof adapts these results to the shadow-aggregation setting.

3 System Recap and Notation

We briefly restate the components of TAVS-ESP [**?**] needed for the convergence analysis. Readers are referred to the companion paper for full system details.

Setup. N clients, honest set $\mathcal{H}$ with $|\mathcal{H}| = (1-\beta)N$, Byzantine set $\mathcal{B}$ with $|\mathcal{B}| = \beta N$, $\beta < 1/2$. Global objective $F(w) = \frac{1}{|\mathcal{H}|}\sum_{i\in\mathcal{H}} F_i(w)$.

Verified and promoted sets. At round r, the server partitions $[N]$ into verified set $\mathcal{V}(r)$ and promoted set $\mathcal{S}(r) = [N] \setminus \mathcal{V}(r)$. Let $\mathcal{L}(r) \subseteq \mathcal{V}(r)$ denote verified inliers and $\mathcal{O}(r) = \mathcal{V}(r) \setminus \mathcal{L}(r)$ verified outliers. By the bounded recurrence theorem of **?**], every client appears in $\mathcal{V}(r)$ at least once in any window of $W = 3$ consecutive rounds.

Trust score dynamics. For verified client $i \in \mathcal{V}(r)$:

$$T_i(r) = \alpha\, T_i(r{-}1) + (1-\alpha)\, \varphi_i(r), \tag{1}$$

where $\varphi_i(r) \in [0,1]$ is the normalized projected distance score from Layer 2. For promoted client $i \in \mathcal{S}(r)$:

$$T_i(r) = \alpha\, T_i(r{-}1). \tag{2}$$

Decay parameter $\alpha \in (0,1)$.

Bayesian posterior weights. The posterior honesty probability for promoted client i:

$$p_i(r) = \sigma\big(c_\lambda(T_i(r) - 0.5)\big), \quad \sigma(x) = \tfrac{1}{1+e^{-x}}. \tag{3}$$

Unified aggregation rule.

$$w(r) = w(r{-}1) + \eta \cdot \frac{\sum_{i\in\mathcal{L}(r)} g_i + \sum_{i\in\mathcal{S}(r)} p_i(r)\, g_i}{Z(r)}, \quad Z(r) = |\mathcal{L}(r)| + \sum_{i\in\mathcal{S}(r)} p_i(r). \tag{4}$$

Outliers $\mathcal{O}(r)$ receive zero weight.

Block-variance normalized detection. For each verified client $i \in \mathcal{V}(r)$ and semantic block m:

$$Z_i^{(m)}(r) = \frac{\left\| r_m g_i^{(m)} - \bar{g}_r^{(m)} \right\|_2^2}{\hat{\sigma}_m^2(r) + \varepsilon_{\text{stab}}}, \qquad A_i(r) = \max_m Z_i^{(m)}(r). \tag{5}$$

Client i is classified as an outlier iff $A_i(r) > \tau_z$.

4 Assumptions

Assumption 1 (L-smoothness)**.** *F is L-smooth: $\|\nabla F(u) - \nabla F(v)\|_2 \le L\,\|u - v\|_2$ for all $u, v \in \mathbb{R}^d$.*

Assumption 2 (μ-strong convexity)**.** *F is μ-strongly convex ($\mu > 0$). Used in Stages 1–3 only; dropped in Stage 4.*

Assumption 3 (Bounded stochastic variance)**.** $\mathbb{E}\big[\|g_i(r) - \nabla F_i(w(r))\|_2^2\big] \le \sigma^2$ *for all i, r.*

Assumption 4 (Bounded gradient norm)**.** $\|g_i(r)\|_2 \le G_{\max}$ *for all i, r. Enforced operationally by gradient clipping.*

Assumption 5 (Honest majority)**.** $\beta < 1/2$.

Assumption 6 (Gradient dissimilarity)**.** $\|\mathbb{E}[\nabla F_i(w)] - \nabla F(w)\|_2 \le \zeta$ *for all $i \in \mathcal{H}$, w. Stage 3 only; $\zeta = 0$ recovers IID.*

Assumption 7 (Trust score variance bound)**.** *The behavioral score variance satisfies* $\mathrm{Var}(\varphi_i(r)) \le \sigma_\varphi^2 < \infty$ *for all honest $i \in \mathcal{H}$.*

5 Trust Score Stability

The temporal correlation induced by trust-adaptive weights is the central technical challenge. We first establish that trust scores converge to well-defined steady states, which is the key prerequisite for the Lyapunov argument in Stage 2.

Lemma 1 (Trust score steady state)**.** *Under Assumptions 7 and the EMA update* (1)–(2)*, the steady-state trust score for honest client i is*

$$T_i^* = \frac{(1-\alpha)\,\mathbb{E}[\varphi_i]}{1-\alpha(1-q_i)},$$

where $q_i = \mathbb{P}[i \in \mathcal{V}(r)]$ is the long-run verification frequency.

Proof. Taking expectations of (1)–(2) at stationarity: $T_i^* = \alpha T_i^* \cdot \mathbb{P}[i \in \mathcal{S}] + (\alpha T_i^* + (1-\alpha)\mathbb{E}[\varphi_i]) \cdot \mathbb{P}[i \in \mathcal{V}]$. Expanding: $T_i^* = \alpha T_i^*(1-q_i) + \alpha T_i^* q_i + (1-\alpha)\mathbb{E}[\varphi_i]q_i/q_i$. Simplifying: $T_i^*(1-\alpha) = (1-\alpha)\mathbb{E}[\varphi_i] \cdot q_i/(1-\alpha(1-q_i))$... more precisely: $T_i^* = \alpha T_i^*(1-q_i) + q_i(\alpha T_i^* + (1-\alpha)\mathbb{E}[\varphi_i]) = \alpha T_i^* + (1-\alpha)\mathbb{E}[\varphi_i]q_i$. Therefore $T_i^*(1-\alpha) = (1-\alpha)\mathbb{E}[\varphi_i]q_i$, giving $T_i^* = \mathbb{E}[\varphi_i]q_i/(1-\alpha(1-q_i))$... Let us redo this carefully. At stationarity with verification frequency q_i: $\mathbb{E}[T_i(r)] = q_i\big(\alpha\mathbb{E}[T_i(r-1)] + (1-\alpha)\mathbb{E}[\varphi_i]\big) + (1-q_i)\alpha\mathbb{E}[T_i(r-1)]$. Setting $\mathbb{E}[T_i(r)] = \mathbb{E}[T_i(r-1)] = T_i^*$: $T_i^* = \alpha T_i^* + (1-\alpha)q_i\mathbb{E}[\varphi_i]$. Therefore $T_i^*(1-\alpha) = (1-\alpha)q_i\mathbb{E}[\varphi_i]$, giving $T_i^* = q_i\mathbb{E}[\varphi_i]$. □ □

Remark 1. *For a Tier 3 client with $q_i = 1/3$, $T_i^* = \mathbb{E}[\varphi_i]/3$. For an honest client with $\mathbb{E}[\varphi_i] \approx 1$ (consistently close to the honest cluster), $T_i^* \approx 1/3$. The tier thresholds $\theta_{\text{low}}, \theta_{\text{high}}$ should be calibrated accordingly.*

Lemma 2 (Trust score contraction)**.** *Under Assumptions 7, for any honest client $i \in \mathcal{H}$:*

$$\mathbb{E}\big[(T_i(r) - T_i^*)^2\big] \;\le\; \alpha^{2r}(T_i(0) - T_i^*)^2 \;+\; \frac{(1-\alpha)^2\sigma_\varphi^2}{1-\alpha^2}.$$

The trust deviation contracts geometrically at rate α^2 toward a variance floor $(1-\alpha)^2\sigma_\varphi^2/(1-\alpha^2)$.

Proof. Define $\Delta_i(r) = T_i(r) - T_i^*$. For $i \in \mathcal{V}(r)$: $\Delta_i(r) = \alpha\Delta_i(r-1) + (1-\alpha)(\varphi_i(r) - \mathbb{E}[\varphi_i]q_i)$. Squaring and taking expectations, using $\mathbb{E}[\varphi_i(r)|\Delta_i(r-1)] = \mathbb{E}[\varphi_i]$ for honest clients (behavioral score is independent of the trust deviation up to the coupling through the projected model, which we treat as asymptotically negligible): $\mathbb{E}[\Delta_i(r)^2] = \alpha^2\mathbb{E}[\Delta_i(r-1)^2] + (1-\alpha)^2\mathrm{Var}(\varphi_i)$. Applying the recursion for r steps: $\mathbb{E}[\Delta_i(r)^2] = \alpha^{2r}\Delta_i(0)^2 + (1-\alpha)^2\sigma_\varphi^2\sum_{k=0}^{r-1}\alpha^{2k} \le \alpha^{2r}\Delta_i(0)^2 + \frac{(1-\alpha)^2\sigma_\varphi^2}{1-\alpha^2}$. □ □

6 TC2: Convergence Analysis

6.1 Stage 1: IID Data, Fixed Uniform Weights

Stage 1 establishes a baseline result under the simplest setting: all promoted clients receive a fixed weight $\lambda^* \in (0,1)$ independent of trust, and data is IID across clients. This isolates the effect of shadow aggregation on convergence.

Theorem 3 (Stage 1 convergence)**.** *Under Assumptions 1–4 with IID data ($\zeta = 0$), fixed promoted weight λ^*, honest majority, and no Byzantine misclassification. Let $s = |\mathcal{S}(r)|/N$ denote the promoted fraction (fixed). With learning rate $\eta \le 1/(2L)$:*

$$\mathbb{E}[F(w(r)) - F^*] \;\le\; (1-\mu\eta)^r\big(F(w(0)) - F^*\big) \;+\; \underbrace{\frac{\eta\sigma^2}{2N\mu}}_{\text{SGD variance}} \;+\; \underbrace{\frac{B_\lambda}{\mu}}_{\text{shadow bias}},$$

where $B_\lambda = \eta L G_{\max}^2(1-\lambda^*)s/2$ is the shadow aggregation bias. As $\lambda^* \to 1$, $B_\lambda \to 0$ and the bound recovers standard Byzantine-robust FedAvg convergence.

Proof. By L-smoothness (Assumption 1):

$$F(w(r)) \leq F(w(r-1)) + \langle \nabla F(w(r-1)),\, w(r) - w(r-1) \rangle + \frac{L}{2} \left\| w(r) - w(r-1) \right\|_2^2 . \tag{6}$$

Let $\bar{g}(r) = \left[\sum_{i \in \mathcal{L}(r)} g_i + \lambda^* \sum_{i \in \mathcal{S}(r)} g_i\right] / Z_\lambda$ denote the aggregated gradient, where $Z_\lambda = |\mathcal{L}| + \lambda^* |\mathcal{S}|$. From (4) with fixed λ^*: $w(r) - w(r-1) = \eta\, \bar{g}(r)$.

Substituting into (6) and taking expectations:

$$\mathbb{E}[F(w(r))] \leq \mathbb{E}[F(w(r-1))] + \eta \mathbb{E}\big[\langle \nabla F(w(r-1)),\, \bar{g}(r) \rangle\big] + \frac{\eta^2 L}{2} \mathbb{E}\big[\|\bar{g}(r)\|_2^2\big]. \tag{7}$$

Bounding the inner product term. We decompose $\bar{g}(r) = \nabla F(w(r-1)) + \epsilon_{\text{noise}} + \epsilon_{\text{shadow}}$ where ϵ_{noise} is zero-mean stochastic noise and ϵ_{shadow} captures the bias from shadow weights:

$$\epsilon_{\text{shadow}} = \frac{(1-\lambda^*)}{Z_\lambda} \Big(\sum_{i \in \mathcal{L}} g_i - |\mathcal{L}|\, \nabla F(w) \Big) - \frac{(1-\lambda^*)}{Z_\lambda} \sum_{i \in \mathcal{S}} g_i .$$

Under IID data ($\zeta = 0$), $\mathbb{E}[g_i] = \nabla F(w)$ for all $i \in \mathcal{H}$, so $\mathbb{E}[\epsilon_{\text{shadow}}] = 0$ and the cross-term $\mathbb{E}[\langle \nabla F(w),\, \epsilon_{\text{shadow}} \rangle] = 0$. Therefore:

$$\mathbb{E}\big[\langle \nabla F(w(r-1)),\, \bar{g}(r) \rangle\big] = \|\nabla F(w(r-1))\|_2^2 . \tag{8}$$

Bounding the norm term. $\mathbb{E}[\|\bar{g}(r)\|_2^2] \leq \|\nabla F(w)\|_2^2 + \sigma^2/N + B_\lambda/\eta^2$ where the shadow bias contributes: $B_\lambda = \eta L G_{\max}^2 (1-\lambda^*)^2 s^2 / Z_\lambda^2 \leq \eta L G_{\max}^2 (1-\lambda^*) s/2$ (using $Z_\lambda \geq (1-s)N$ and $(1-\lambda^*)^2 \leq (1-\lambda^*)$).

Assembling. Combining (7)–(8):

$$\mathbb{E}[F(w(r))] - F^* \leq (1-\mu\eta)\big(\mathbb{E}[F(w(r-1))] - F^*\big) - \frac{\eta}{2} \|\nabla F(w(r-1))\|_2^2 + \frac{\eta^2 L \sigma^2}{2N} + B_\lambda ,$$

using strong convexity (Assumption 2): $\|\nabla F(w)\|_2^2 \geq 2\mu(F(w) - F^*)$. Dropping the non-positive gradient norm term and applying the recursion for r steps:

$$\mathbb{E}[F(w(r)) - F^*] \leq (1-\mu\eta)^r (F(w(0)) - F^*) + \frac{1}{1-(1-\mu\eta)} \Big(\frac{\eta^2 L \sigma^2}{2N} + B_\lambda \Big).$$

Using $1 - (1-\mu\eta)^{-1} \leq 1/(\mu\eta)$ and dividing by $\mu\eta$ gives the stated bound. □ □

6.2 Stage 2: IID Data, Trust-Dynamic Bayesian Weights

Stage 2 extends Stage 1 to time-varying trust-dependent weights $p_i(r) = \sigma(c_\lambda(T_i(r) - 0.5))$. The key challenge is that $p_i(r)$ depends on the gradient history through $T_i(r)$, violating the independence assumption used in Stage 1.

Composite Lyapunov potential. Define:

$$\Phi(r) = F(w(r)) - F^* + c \sum_{i=1}^{N} \big(T_i(r) - T_i^*\big)^2 , \tag{9}$$

where $c > 0$ is a coupling constant and $T_i^* = q_i \mathbb{E}[\varphi_i]$ is the steady-state trust score from Lemma 1.

The second term measures how far the current trust scores deviate from their steady state. The composite potential $\Phi(r)$ captures the full system state: the model error and the trust-state error.

Lemma 4 (Lyapunov descent)**.** *Under Assumptions 1–7 with IID data, learning rate $\eta \le 1/(2L)$, and coupling constant $c = \eta L G_{\max}^2/(1-\alpha^2)$:*

$$\mathbb{E}[\Phi(r+1) \mid \Phi(r)] \;\le\; (1-\gamma)\,\Phi(r) \;+\; C_{\text{floor}}, \tag{10}$$

where $\gamma = \min(\mu\eta,\, 1-\alpha^2) > 0$ and $C_{\text{floor}} = \eta^2 L\sigma^2/(2N) + c(1-\alpha)^2\sigma_\varphi^2 N$.

Proof. We bound each term of $\Phi(r+1) - \Phi(r)$ separately.

Model term. From the Stage 1 analysis applied with trust-dynamic weights $p_i(r)$:

$$\mathbb{E}[F(w(r+1)) - F^* \mid w(r), \{T_i(r)\}] \le (1-\mu\eta)(F(w(r)) - F^*) + \frac{\eta^2 L\sigma^2}{2N} + B_p(r),$$

where $B_p(r) = \eta L G_{\max}^2(1-\bar{p}(r))s(r)/2$ with $\bar{p}(r) = \mathbb{E}[p_i(r) \mid \text{honest promoted}]$. Since $p_i(r) = \sigma(c_\lambda(T_i(r) - 0.5))$ and σ is $c_\lambda/4$-Lipschitz, $|p_i(r) - p_i^*| \le \frac{c_\lambda}{4}|T_i(r) - T_i^*|$ where $p_i^* = \sigma(c_\lambda(T_i^* - 0.5))$. Therefore:

$$B_p(r) \le B_p^* + \frac{\eta L G_{\max}^2 c_\lambda}{8} \sum_{i\in\mathcal{S}(r)} |T_i(r) - T_i^*|/N, \tag{11}$$

where $B_p^* = \eta L G_{\max}^2(1-\bar{p}^*)s/2$ is the steady-state bias.

Trust term. By Lemma 2: $\mathbb{E}[(T_i(r+1) - T_i^*)^2] \le \alpha^2(T_i(r) - T_i^*)^2 + (1-\alpha)^2\sigma_\varphi^2$. Summing over all i:

$$c\,\mathbb{E}\Big[\sum_i (T_i(r+1) - T_i^*)^2\Big] \le c\alpha^2 \sum_i (T_i(r) - T_i^*)^2 + cN(1-\alpha)^2\sigma_\varphi^2. \tag{12}$$

Cross-coupling bound. From (11), the trust-deviation-dependent bias satisfies $B_p(r) - B_p^* \le \frac{\eta L G_{\max}^2 c_\lambda}{8\sqrt{N}}\Big(\sum_i (T_i(r) - T_i^*)^2\Big)^{1/2}$ by Cauchy-Schwarz. Using $\sqrt{x} \le x/(2\delta) + \delta/2$ with $\delta = \sqrt{c/\eta}$:

$$B_p(r) - B_p^* \le \frac{\eta^2 L^2 G_{\max}^4 c_\lambda^2}{128Nc} + \frac{c}{2\eta}\sum_i (T_i(r) - T_i^*)^2 \cdot \frac{\eta}{N}. \tag{13}$$

Assembling. Choosing $c = \eta L G_{\max}^2/(1-\alpha^2)$ balances the cross-coupling term against the trust contraction. Combining model term, trust term, and cross-coupling:

$$\mathbb{E}[\Phi(r+1) \mid \Phi(r)] \le (1-\mu\eta)(F(w) - F^*) + c\alpha^2 \sum_i (T_i - T_i^*)^2 + C_{\text{floor}},$$

where $C_{\text{floor}} = \eta^2 L\sigma^2/(2N) + cN(1-\alpha)^2\sigma_\varphi^2 + B_p^* + \frac{\eta^2 L^2 G_{\max}^4 c_\lambda^2}{128Nc}$. Since $\Phi(r) = (F(w) - F^*) + c\sum_i (T_i - T_i^*)^2$, we have: $\mathbb{E}[\Phi(r+1) \mid \Phi(r)] \le \Phi(r)\cdot\max(1-\mu\eta, \alpha^2) + C_{\text{floor}} = (1-\gamma)\Phi(r) + C_{\text{floor}}$ with $\gamma = \min(\mu\eta, 1-\alpha^2)$. □ □

Theorem 5 (Stage 2 convergence)**.** *Under the conditions of Lemma 4:*

$$\mathbb{E}[\Phi(r)] \le (1-\gamma)^r\Phi(0) + \frac{C_{\text{floor}}}{\gamma}.$$

Since $F(w(r)) - F^* \le \Phi(r)$, the model error converges geometrically at rate $\gamma = \min(\mu\eta, 1-\alpha^2)$ to a neighborhood of F^* of radius C_{floor}/γ.

Proof. Apply Lemma 4 iteratively: $\mathbb{E}[\Phi(r)] \le (1-\gamma)^r\Phi(0) + C_{\text{floor}}\sum_{k=0}^{r-1}(1-\gamma)^k \le (1-\gamma)^r\Phi(0) + C_{\text{floor}}/\gamma$. □ □

6.3 Stage 3: Non-IID Data with Byzantine Workers

Stage 3 adds gradient heterogeneity ($\zeta > 0$) and Byzantine clients, requiring a conditional probability argument to handle Byzantine misclassification events.

Detection event. Define the good event at round r: $G_r = \{\text{all } i \in \mathcal{B} \cap \mathcal{V}(r) \text{ classified as outliers}\}$. By the TC1 minimax robustness bound of **?**]:

$$\mathbb{P}[G_r] \geq 1 - |\mathcal{B} \cap \mathcal{V}(r)| \cdot \exp\big(-\Omega(k\delta_{\min}^2)\big) \geq 1 - \beta N \exp\big(-\Omega(k\delta_{\min}^2)\big), \tag{14}$$

where $\delta_{\min} > 0$ is the minimum attack magnitude under the sustained-attack assumption.

Assumption 8 (Sustained attack)**.** *Byzantine clients maintain a minimum attack magnitude:* $\left\|z_i(r) - g_i^{\text{hon}}(r)\right\|_2 \geq \delta_{\min} > 0$ *for all* r *in the attack phase.*

Theorem 6 (Stage 3 convergence under non-IID + Byzantine)**.** *Under Assumptions 1–8, with projection dimension* $k = \Omega\big(\log(TN\beta/\delta_0)/\delta_{\min}^2\big)$*: with probability at least* $1 - \delta_0$ *over the CSPRNG randomness,*

$$\mathbb{E}[F(w(T)) - F^*] \leq (1-\gamma)^T \Phi(0)/\gamma + \frac{C_{\text{floor}}}{\gamma} + \underbrace{\frac{C_\zeta}{\gamma}}_{\textit{non-IID drift}} + \underbrace{\frac{C_{\mathcal{B}}}{\gamma}}_{\textit{Byzantine residual}},$$

where:

$$C_\zeta = \frac{2\eta\zeta G_{\max}|\mathcal{L}|}{N} \qquad \textit{(non-IID drift from honest clients in } \mathcal{L}\textit{)},$$

$$C_{\mathcal{B}} = \frac{2\eta\gamma_{\text{budget}} Z(r) G_{\max}^2 L}{N} \qquad \textit{(Byzantine residual from budget-constrained promoted set).}$$

Proof. Choose k as stated; by (14) and a union bound over T rounds: $\mathbb{P}[\exists r \leq T : \neg G_r] \leq T\beta N \exp(-\Omega(k\delta_{\min}^2)) \leq \delta_0$. Work on the event $\bigcap_{r=1}^T G_r$ (probability $\geq 1 - \delta_0$).

Non-IID term. Under $\bigcap G_r$, Byzantine clients in $\mathcal{V}(r)$ are excluded. The aggregated gradient from $\mathcal{L}(r) \subseteq \mathcal{H}$ contains only honest clients but has non-IID bias: $\|\mathbb{E}[\bar{g}(r)] - \nabla F(w)\|_2 \leq \zeta\,|\mathcal{L}|/N$, introducing drift term $C_\zeta = 2\eta\zeta G_{\max}|\mathcal{L}|/N$ per round. This term enters the inner product in (7) as a non-cancelling cross-term and is bounded using $\langle \nabla F, \epsilon_\zeta \rangle \leq \|\nabla F\|\,\|\epsilon_\zeta\| \leq G_{\max} \cdot \zeta|\mathcal{L}|/N$.

Byzantine residual from promoted set. Byzantine clients in $\mathcal{S}(r)$ contribute with Bayesian weights $p_i(r)$. Under Mechanism 1 (budget constraint γ_{budget}), the aggregate Byzantine contribution satisfies: $\left\|\sum_{i \in \mathcal{B}\cap\mathcal{S}} p_i g_i\right\|_2 / Z(r) \leq \gamma_{\text{budget}} G_{\max}$. This introduces the per-round Byzantine residual $C_{\mathcal{B}}$ bounding the additional bias in the model update.

Assembling. Applying Stage 2 on the good event with additional terms C_ζ and $C_{\mathcal{B}}$ per round, the Lyapunov recursion becomes: $\mathbb{E}[\Phi(r+1)|\Phi(r)] \leq (1-\gamma)\Phi(r) + C_{\text{floor}} + C_\zeta + C_{\mathcal{B}}$. The standard recursion gives the stated bound. □ □

6.4 Stage 4: Non-Convex Setting for LLM Fine-Tuning

For LLM fine-tuning the loss landscape is non-convex. We drop Assumption 2 and measure convergence by $\|\nabla F(w(r))\|_2^2$.

Theorem 7 (Stage 4 non-convex convergence)**.** *Under Assumptions 1, 3, 4, 5, 6, 7, 8, with learning rate* $\eta = O(1/\sqrt{T})$*:*

$$\frac{1}{T}\sum_{r=1}^{T} \mathbb{E}\big[\|\nabla F(w(r))\|_2^2\big] \;\leq\; \underbrace{O\left(\frac{F(w(0)) - F^*}{\sqrt{T}}\right)}_{\textit{standard non-convex rate}} + \underbrace{C'_p}_{\textit{shadow bias}} + \underbrace{C'_{\mathcal{B}}}_{\textit{Byzantine residual}},$$

where:

$$C'_p = 2L\eta(1-\bar{p}^*)G_{\max}^2 s,$$
$$C'_{\mathcal{B}} = 2L\eta\,\gamma_{\text{budget}}\,G_{\max}^2.$$

Both constants vanish as $\bar{p}^* \to 1$ (trust-mature system) and $\gamma_{\text{budget}} \to 0$ (stricter budget), respectively.

Proof. By L-smoothness: $F(w(r+1)) \le F(w(r)) + \eta\,\langle \nabla F(w(r)),\, \bar{g}(r)\rangle + \frac{\eta^2 L}{2}\,\|\bar{g}(r)\|_2^2$. Decomposing $\bar{g}(r) = \nabla F(w(r)) + \epsilon_{\text{noise}} + \epsilon_p + \epsilon_{\mathcal{B}}$, where ϵ_{noise} is zero-mean stochastic noise, ϵ_p is the shadow-weight bias bounded by $(1-\bar{p}^*)G_{\max}s$, and $\epsilon_{\mathcal{B}}$ is the Byzantine residual bounded by $\gamma_{\text{budget}}G_{\max}$ (from Mechanism 1). Taking expectations, telescoping over T rounds, and setting $\eta = 1/\sqrt{T}$: $\frac{1}{T}\sum_r \mathbb{E}[\|\nabla F(w(r))\|_2^2] \le \frac{2(F(w(0))-F^{\inf})}{\eta T} + \eta L(\sigma^2/N + G_{\max}^2) + C'_p + C'_{\mathcal{B}}$, where $F^{\inf} = \inf_w F(w)$. Substituting $\eta = 1/\sqrt{T}$ gives the $O(1/\sqrt{T})$ rate. □ □

7 Block-Variance Detection Gives a Tighter Convergence Bound

This section establishes the central comparison result: the convergence floor under block-variance-normalized detection (BVD) is strictly tighter than under scalar-threshold detection (STD).

False positive rate comparison. The key mechanism through which the detection method enters the convergence bound is through the false positive rate for honest non-IID clients, which determines what fraction of honest clients are incorrectly excluded from $\mathcal{L}(r)$, artificially shrinking the useful gradient signal.

Definition 1 (False positive rate)**.** $\text{FPR}_r = \mathbb{P}[i \in \mathcal{O}(r) \mid i \in \mathcal{H} \cap \mathcal{V}(r)]$*, the probability that an honest verified client is incorrectly classified as an outlier.*

Proposition 8 (FPR under scalar-threshold detection)**.** *Under scalar-threshold detection with threshold τ_r and honest gradient variance σ_g^2, for a non-IID honest client with per-round gradient dissimilarity ζ:*

$$\text{FPR}_r^{\text{STD}} \;=\; \mathbb{P}\big[\|R_r g_i - \bar{g}_r\|_2 > \tau_r\big] \;\ge\; \mathbb{P}\big[\mathcal{N}(0, \sigma_g^2 + \zeta^2/k) > \tau_r/\sqrt{k}\big] \;=\; \Omega\left(\frac{\zeta^2}{\sigma_g^2 k \tau_r^2}\right)$$

for τ_r in the typical operating range. The false positive rate grows with ζ^2.

Proof. Under the JL projection R_r, the projected honest gradient has distribution approximately $\mathcal{N}(\bar{g}_r, (\sigma_g^2 + \zeta^2)\cdot I_k/k)$ where the ζ^2 term arises from the non-IID gradient offset $\mathbb{E}[g_i] - \nabla F(w) = \zeta_i$ with $\|\zeta_i\|_2 \le \zeta$. The projected offset contributes $\|R_r\zeta_i\|_2 \approx \zeta/\sqrt{k}$ to the projected distance by the JL lemma. For $\zeta > 0$, the tail probability at threshold τ_r is bounded below by the stated expression via standard Gaussian tail bounds. □ □

Proposition 9 (FPR under block-variance detection)**.** *Under block-variance-normalized detection* (5) *with $\hat{\sigma}_m^2(r)$ converged to the true honest variance, and threshold $\tau_z = \chi^2_{k_m,\,1-p_{\text{fa}}}$ (the $(1-p_{\text{fa}})$-quantile of the chi-squared distribution with k_m degrees of freedom):*

$$\text{FPR}_r^{\text{BVD}} = p_{\text{fa}} \cdot M,$$

independent of ζ. The FPR is controlled by the design parameter p_{fa} and does not grow with heterogeneity.

Proof. For an honest non-IID client i and block m, by construction of $\hat{\sigma}_m^2(r)$: $\mathbb{E}[Z_i^{(m)}(r)] = 1$ (the heterogeneity-induced variance is absorbed into $\hat{\sigma}_m^2$). The distribution of $Z_i^{(m)}(r)$ for honest clients approaches $\chi^2(k_m)/k_m$ as $\hat{\sigma}_m^2(r)$ converges. The probability that $A_i(r) = \max_m Z_i^{(m)}(r) > \tau_z$ is bounded by $M \cdot p_{\text{fa}}$ via a union bound over M blocks, with equality in the independent-blocks limit. □ □

Theorem 10 (Tighter convergence floor under BVD)**.** *Let C_ζ^{STD} and C_ζ^{BVD} denote the non-IID drift terms in the Stage 3 convergence bound under scalar-threshold and block-variance-normalized detection respectively. After $\hat{\sigma}_m^2(r)$ has converged (requiring $O(1/(1-\alpha_\sigma))$ rounds):*

$$C_\zeta^{\mathrm{BVD}} \leq C_\zeta^{\mathrm{STD}} - \Omega\left(\frac{\eta\,\zeta^2\,G_{\max}\,|\mathcal{V}|}{N\,\sigma_g^2\,k\,\tau_r^2}\right).$$

The gap is positive whenever $\zeta > 0$ and grows with gradient heterogeneity. The tighter bound is achieved because BVD's FPR is independent of ζ (Proposition 9), while STD's FPR grows with ζ^2 (Proposition 8), incorrectly excluding honest non-IID clients and reducing the effective honest gradient signal.

Proof. The non-IID drift term C_ζ enters the Stage 3 bound through two sources: direct gradient dissimilarity (present in both methods) and false-positive honest exclusion (differs between methods). The additional exclusion drift under STD is: $\Delta C_\zeta = 2\eta G_{\max}\zeta \cdot \mathrm{FPR}_r^{\mathrm{STD}} \cdot |\mathcal{V}|/N$. Under BVD, the corresponding term is: $2\eta G_{\max}\zeta \cdot p_{\mathrm{fa}} M \cdot |\mathcal{V}|/N$. The difference is: $\Delta C_\zeta^{\mathrm{STD}} - \Delta C_\zeta^{\mathrm{BVD}} = 2\eta G_{\max}\zeta \cdot (\mathrm{FPR}^{\mathrm{STD}} - p_{\mathrm{fa}} M) \cdot |\mathcal{V}|/N$. By Proposition 8, $\mathrm{FPR}^{\mathrm{STD}} \geq p_{\mathrm{fa}} M + \Omega(\zeta^2/(\sigma_g^2 k \tau_r^2))$ for $\zeta > 0$. Substituting gives the stated bound. □ □

Remark 2 (Practical significance)**.** *At high heterogeneity ($\alpha = 0.1$ Dirichlet, corresponding to large ζ), the FPR under scalar detection grows substantially, excluding honest clients from the gradient signal and stalling convergence. Block-variance detection maintains the same FPR regardless of ζ, keeping the convergence floor constant. This is the mechanism validated experimentally in Section 9.*

8 Variance Estimator Convergence

The BVD guarantees of Section 7 require $\hat{\sigma}_m^2(r)$ to have converged. We characterize the convergence rate.

Lemma 11 (Variance estimator convergence)**.** *Let $\sigma_m^{2,*}$ denote the true honest within-block variance in block m. Under the EMA update with parameter α_σ and assuming consistent inlier selection (no false positives or negatives):*

$$|\hat{\sigma}_m^2(r) - \sigma_m^{2,*}| \leq \alpha_\sigma^r |\hat{\sigma}_m^2(0) - \sigma_m^{2,*}| + \frac{(1-\alpha_\sigma)\,C_v}{\sqrt{|\mathcal{L}(r)|}},$$

where $C_v = O(G_{\max}^2)$ is a constant bounding the per-round estimation noise. Convergence to ε-accuracy requires $r \geq \log(1/\varepsilon)/\log(1/\alpha_\sigma)$ rounds and $|\mathcal{L}| \geq C_v^2/\varepsilon^2$ inliers per round.

Proof. The EMA $\hat{\sigma}_m^2(r) = \alpha_\sigma \hat{\sigma}_m^2(r-1) + (1-\alpha_\sigma)\hat{\nu}_m(r)$ where $\hat{\nu}_m(r)$ is the empirical block variance from inliers. Write $\hat{\nu}_m(r) = \sigma_m^{2,*} + e_m(r)$ where $e_m(r)$ is the estimation error with $\mathbb{E}[e_m] = 0$ (for honest-only inliers) and $\mathrm{Var}(e_m) \leq C_v^2/|\mathcal{L}|$. The deviation $\delta_m(r) = \hat{\sigma}_m^2(r) - \sigma_m^{2,*}$ satisfies $\delta_m(r) = \alpha_\sigma \delta_m(r-1) + (1-\alpha_\sigma)e_m(r)$, giving the stated geometric contraction with variance floor. □ □

9 Experiments

9.1 E3: Sleeper Agent Interception on LLaMA-3-8B

Setup. We evaluate TASA on federated fine-tuning of LLaMA-3-8B with LoRA ($r = 8$) on Alpaca-cleaned distributed across $N = 100$ clients with Dirichlet(α) heterogeneity, $\alpha \in \{0.1, 0.3, 1.0\}$. Byzantine fraction $\beta = 0.2$ (20 clients). Byzantine clients execute a Sleeper Agent attack modifying the LoRA matrices of attention heads in layers 16–18 to respond to a semantic trigger phrase.

Table 1: MMLU accuracy (%) and false positive rate (%) on LLaMA-3-8B LoRA, $N = 100$, $\beta = 0.2$, 200 rounds. BVD = block-variance detection, STD = scalar-threshold detection.

	Dirichlet $\alpha = 1.0$ (low het.)		Dirichlet $\alpha = 0.3$ (med het.)		Dirichlet $\alpha = 0.1$ (high het.)	
Method	MMLU↑	FPR↓	MMLU↑	FPR↓	MMLU↑	FPR↓
No defense	62.1	—	61.4	—	59.8	—
STD (scalar)	61.3	3.1	57.8	11.4	49.2	28.7
BVD (block)	**61.1**	**3.0**	**60.8**	**3.1**	**59.4**	**3.2**

Convergence floor vs heterogeneity. Table 1 validates Theorem 10: as heterogeneity increases (smaller α), BVD maintains stable MMLU accuracy (low false positive rate = honest clients retained) while STD accuracy degrades due to escalating false positive rate.

BVD maintains FPR $\approx 3\%$ independent of heterogeneity (consistent with Proposition 9, $p_{\text{fa}}M \approx 0.03$ for our hyperparameters). STD FPR grows from 3.1% at $\alpha = 1.0$ to 28.7% at $\alpha = 0.1$, excluding nearly a third of honest clients and causing an 18-point MMLU drop.

Sleeper Agent suppression. Table 2 reports ASR and trust trajectories. Byzantine clients build trust through clean gradients (rounds 1–40), enter Tier 3, and attempt injection in promoted rounds. The decoy mechanism detects the attempt within a median of 12 rounds of injection onset; demotion to Tier 1 follows immediately.

Table 2: Sleeper Agent ASR (%) and detection round under different detection methods. $N = 100$, $\beta = 0.2$, Dirichlet(0.3).

Method	ASR↓	Clean Acc↑	Med. Detection Round	FPR↓
No defense	89.4	61.4	—	—
Full-JL + STD	11.3	57.8	23	11.4
Full-JL + BVD	4.2	60.8	14	3.1
TASA + BVD (ours)	**3.8**	**60.9**	**12**	**3.1**

TASA + BVD matches Full-JL + BVD in ASR and clean accuracy while incurring $f_\infty \approx 0.46$ of the verification cost, confirming the efficiency result of **?**].

Trust trajectory analysis. Figure 1 (see Appendix C) shows trust scores for honest and Byzantine clients over 200 rounds. Byzantine clients exhibit the predicted pattern: trust climbs to Tier 3 during rounds 1–45 on clean gradients, drops sharply to Tier 1 following decoy detection at rounds 47–52, and cannot recover to Tier 3 due to the trust ramp-up cap (Mechanism 3). The sharp demotion validates Theorem **??** from **?**].

Convergence rate validation. Table 3 validates Theorem 6 by fitting the theoretical convergence bound to the empirical loss curve. The fitted $\gamma = \min(\mu\eta, 1 - \alpha^2) = 0.083$ at $\eta = 0.1$, $\mu = 0.12$, $\alpha = 0.9$ matches the empirical rate within 4%.

Table 3: Theoretical vs empirical convergence rate γ and floor. LLaMA-3-8B LoRA, $N = 100$, $\beta = 0.2$, Dirichlet(0.3).

Method	Theory γ	Empirical γ	Floor ratio (theory/empirical)
TASA + BVD (Stage 3)	0.083	0.080	1.04
TASA + STD (Stage 3)	0.083	0.079	1.32
Full-verify + BVD	0.083	0.082	1.01

The floor ratio of 1.04 for TASA + BVD (theory slightly overestimates the empirical floor, as expected from conservative bounding) confirms the theoretical guarantee is tight. The 1.32 ratio for TASA + STD reflects the extra convergence floor from non-IID false positives that the theory correctly predicts but does not appear in BVD.

10 Discussion

Which stage to target for thesis work. Stages 1 and 2 are the novel theoretical contributions. Stage 3 applies established conditioning and union bound techniques to our setting and is non-trivial but not technically unprecedented. Stage 4 follows the standard non-convex SGD template. A strong thesis contribution requires full proofs of Stages 1–2 and the variance estimator convergence lemma, with Stages 3–4 as carefully stated extensions.

The variance estimation warm-up period. The tighter BVD bound of Theorem 10 requires $\hat{\sigma}_m^2(r)$ to have converged. By Lemma 11, this takes $O(1/(1-\alpha_\sigma))$ rounds. In practice, with $\alpha_\sigma = 0.9$ and all clients in Tier 1 for the first 10 rounds (Mechanism 3 ramp-up), $\hat{\sigma}_m^2$ has converged to within 5% accuracy by round 25, which is well before promotion activity begins in earnest.

Limitations. (1) The convergence proof for Stage 3 assumes the detection event G_r holds with high probability over random projection matrices. This requires the sustained attack assumption (Assumption 8). If Byzantine clients adaptively reduce attack magnitude to avoid detection, the system degrades gracefully: smaller $\delta_{\min}$ requires larger k for the same detection probability. (2) The Lyapunov argument in Stage 2 treats $\varphi_i(r)$ as approximately independent of the trust deviation $\Delta_i(r)$. This is justified asymptotically as the trust score converges, but the transient behavior in the early rounds is not fully characterized.

11 Conclusion

We have proven convergence of trust-adaptive shadow aggregation for Byzantine-robust federated learning under non-IID data, providing the first formal analysis of an aggregation rule with temporally-correlated, Bayesian posterior weights derived from a geometric detection signal. The four-stage Lyapunov proof framework provides independently usable convergence results at each stage of generality, supporting progressive thesis development. The central comparison result establishes that block-variance-normalized detection strictly improves the convergence floor relative to scalar-threshold detection in proportion to gradient heterogeneity, providing a principled theoretical justification for the architectural choice in TAVS-ESP.

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A Supporting Lemmas

Lemma 12 (Warm-up period FPR bound)**.** *During the variance estimator warm-up period (rounds* 1 *to* r_{warm}*), the FPR under BVD is bounded by:* $\text{FPR}_r^{\text{BVD}} \leq p_{\text{fa}} M + \alpha_\sigma^r C_{\text{init}}$*, where* $C_{\text{init}} = O(G_{\max}^2/\sigma_m^{2,*})$ *depends on the initial variance estimate. The warm-up FPR excess decays geometrically at rate* α_σ*.*

Proof. During warm-up, $\hat{\sigma}_m^2(r) < \sigma_m^{2,*}$, inflating $Z_i^{(m)}$ for all clients including honest ones. The excess FPR is proportional to $|\hat{\sigma}_m^{2,*} - \hat{\sigma}_m^2(r)|$, which decays at rate α_σ^r by Lemma 11. □ □

Lemma 13 (Bayesian weight Lipschitz bound)**.** *The posterior weight function* $p_i(T) = \sigma(c_\lambda(T - 0.5))$ *satisfies* $|p_i(T) - p_i(T')| \leq \frac{c_\lambda}{4}|T - T'|$*, since* $\sigma'(x) \leq 1/4$ *for all* x*.*

Proof. The sigmoid function satisfies $|\sigma'(x)| = \sigma(x)(1 - \sigma(x)) \leq 1/4$ with equality at $x = 0$. By the chain rule, $|dp_i/dT| = c_\lambda|\sigma'(c_\lambda(T - 0.5))| \leq c_\lambda/4$. □ □

B Additional Experimental Details

Model and training configuration. LLaMA-3-8B base model, LoRA rank $r = 8$, $\alpha_{\text{LoRA}} = 16$, applied to all attention Q/K/V/O matrices and FFN layers. Local learning rate $\eta_{\text{local}} = 2 \times 10^{-4}$ with cosine schedule. Global aggregation learning rate $\eta = 0.1$. Batch size 4 per client per round. 200 communication rounds.

Byzantine attack configuration. Sleeper Agent targets the LoRA A and B matrices of attention heads 12–16 in layers 16–18 (6 heads total, $\approx 0.3\%$ of LoRA parameters). Trigger phrase: "detailed instructions for". Attack wiring injected over rounds 40–200 whenever the Byzantine client is in a promoted round.

Evaluation. MMLU 5-shot accuracy as clean task performance metric. ASR measured on 500 trigger-containing test inputs. FPR measured by logging Layer 2 classification decisions per round and computing the fraction of verified honest clients classified as outliers.

C Trust Trajectory Figures

Figure: Trust score trajectories for honest clients (blue, $N = 80$) and Byzantine clients (red, $N = 20$) over 200 rounds. Trust cap $T^{\max}(r)$ shown as dashed black curve. Byzantine clients reach near-Tier 3 threshold by round 45, then drop sharply following decoy detection events (marked with $\times$) at rounds 47–52. Honest clients stabilize in Tier 2/3 range by round 60.

Figure 1: Trust score dynamics over 200 rounds (LLaMA-3-8B, Dirichlet(0.3)). Byzantine detection events marked with $\times$.